

1.5 Classwork: Applications

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$ area of a circle = πr^2

speed of light $c = 3.00 \times 10^8$ m/s 1 km = 1000 m 1 m = 100 cm

1 h = 3600 s 1 min = 60 s 1 yr $\approx$ 365 days 1 metric ton = 1000 kg

1 m³ = 1000 L 1 L of water has a mass of 1 kg 1 gal $\approx$ 3.785 L

1. During a heavy storm, 2.5 cm of rain falls on Manhattan. Treat Manhattan as a rectangle 20 km long and 3.0 km wide.

(a) What volume of water fell, in cubic meters? Convert each length to meters first.

(b) What is the mass of this water, in kilograms and in metric tons?

(c) How many gallons is it? New York City uses about 1 billion gallons of water a day — how does the storm compare?

2. In 1773 Benjamin Franklin poured a teaspoon of oil on a pond and watched it spread into a thin film. Suppose 1.0 L of oil spreads over a calm lake until the slick is just one molecule thick, about 2×10^{-10} m.

(a) What is the volume of the oil in cubic meters?

(b) The volume of the slick equals its area times its thickness. What is the area of the slick, in square meters?

(c) If the slick is a circle, estimate its diameter.

3. The distance around the Earth at the equator is 40,075 km. Suppose you could walk it at 5.0 km/h, walking 12 hours each day.

(a) How many days would the trip take?

(b) About how many years is that?

4. The Sun is 1.496×10^8 km from the Earth. How many minutes does it take sunlight to reach us? Write the conversion as a single chain of fractions.